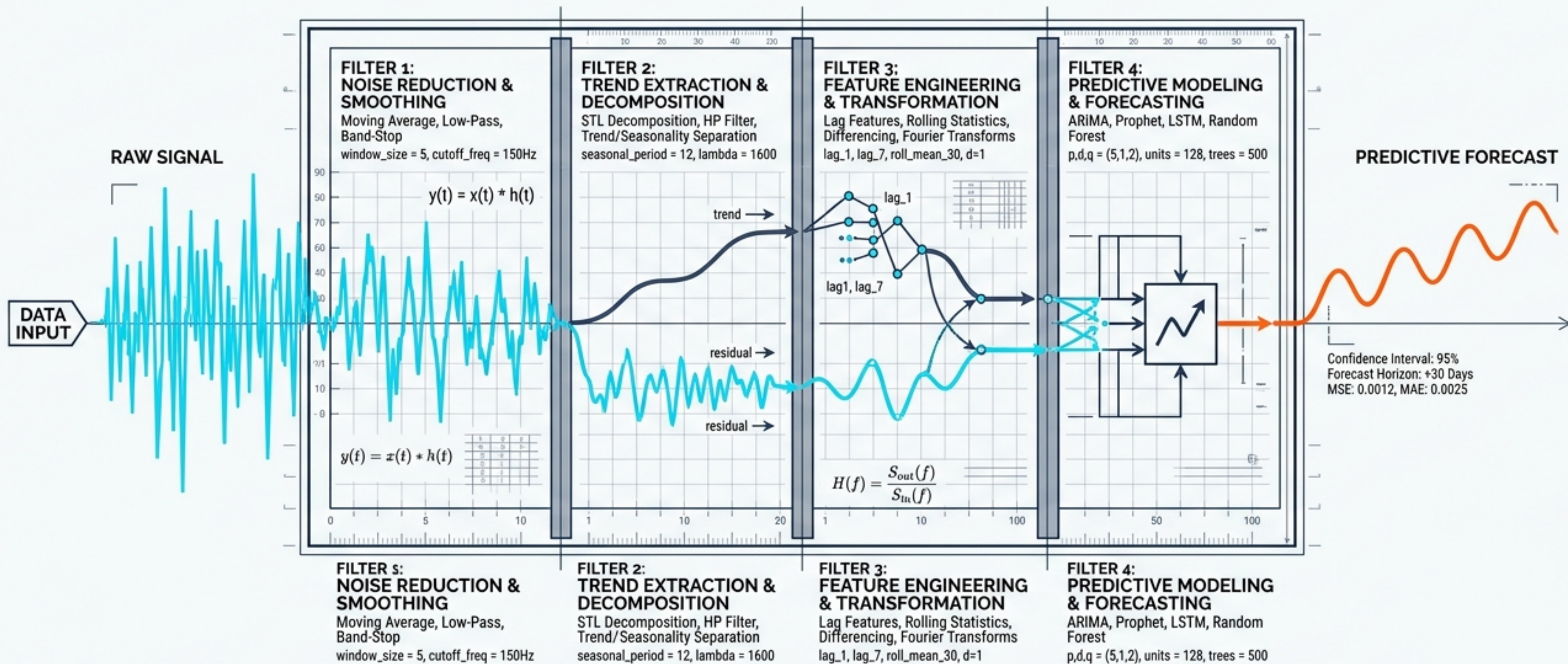


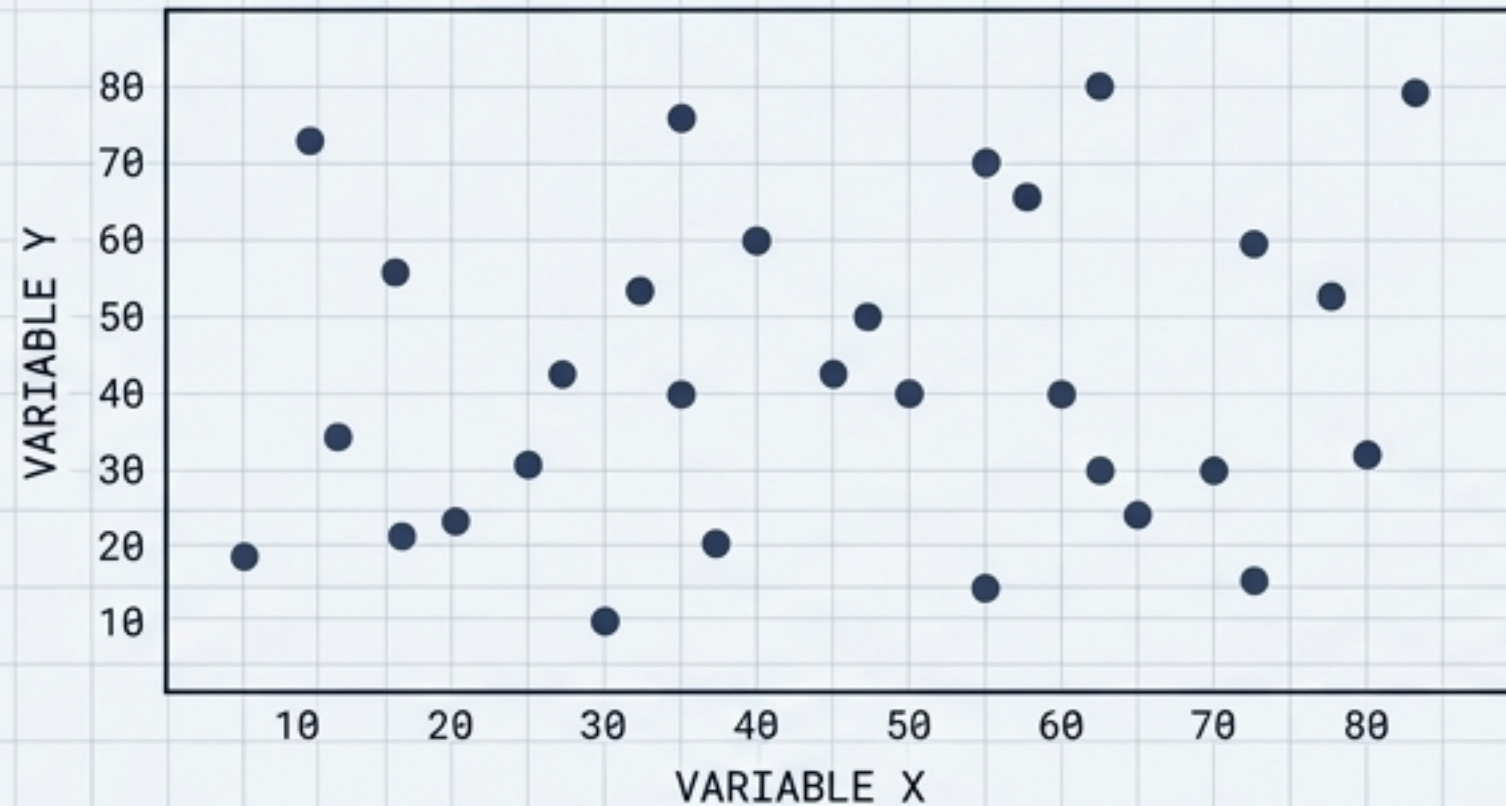
THE TIME SERIES BLUEPRINT

FROM RAW SIGNAL TO PREDICTIVE FORECAST



THE TIME SERIES CHALLENGE: WHY TIME BREAKS TRADITIONAL MATH

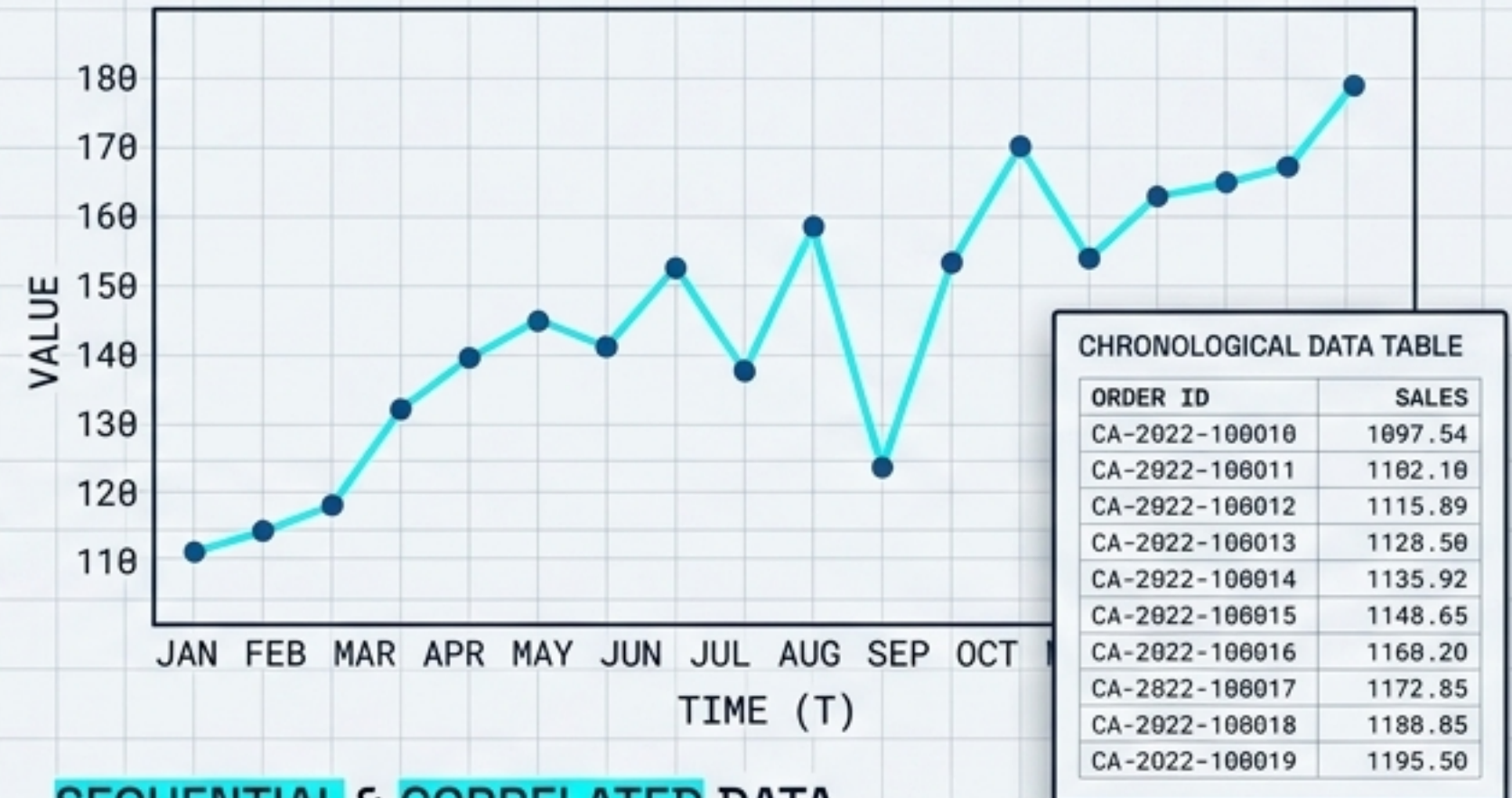
THE TRADITIONAL ML ASSUMPTION



CROSS-SECTIONAL DATA

Variables are **independent**. What happens today has no mathematical bearing on what happens tomorrow.

THE TIME SERIES REALITY



SEQUENTIAL & CORRELATED DATA

Data points are inextricably **linked**. The **past** directly influences the **present** (e.g., today's high sales predict tomorrow's inventory drop).

FORECASTING DIMENSIONS

UNIVARIATE

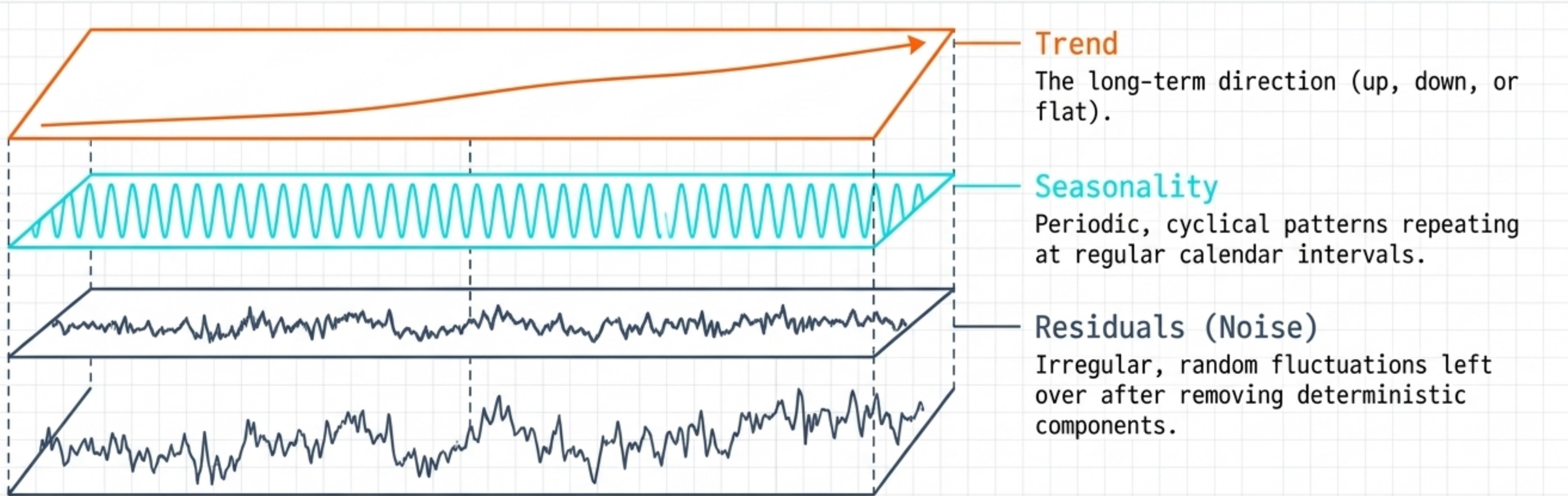
Predicting a single variable based entirely on its own past.

MULTIVARIATE / PANEL

Predicting multiple related, overlapping series that influence each other.

DECONSTRUCTING THE WAVE: TIME SERIES DECOMPOSITION

What is actually inside a messy line?



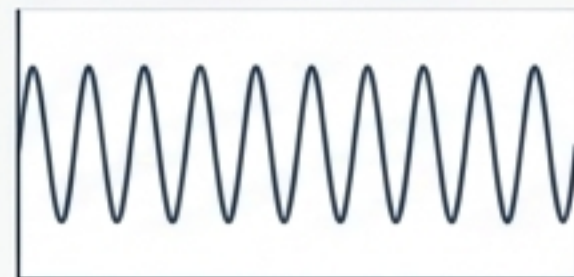
THE MATH OF COMBINATION

Additive Model

$$y_t = T_t + S_t + R_t$$

Constant amplitude. Trend and seasonality are independent.

CONSTANT AMPLITUDE

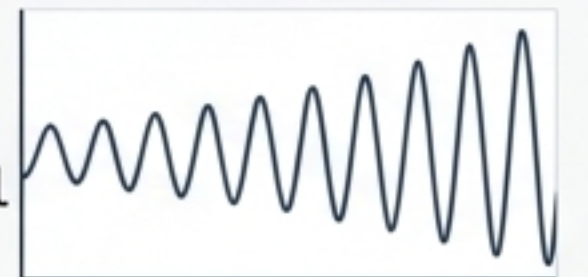


Multiplicative Model

$$y_t = T_t \times S_t \times R_t$$

Growing amplitude. Seasonal and residual magnitudes scale with the trend.

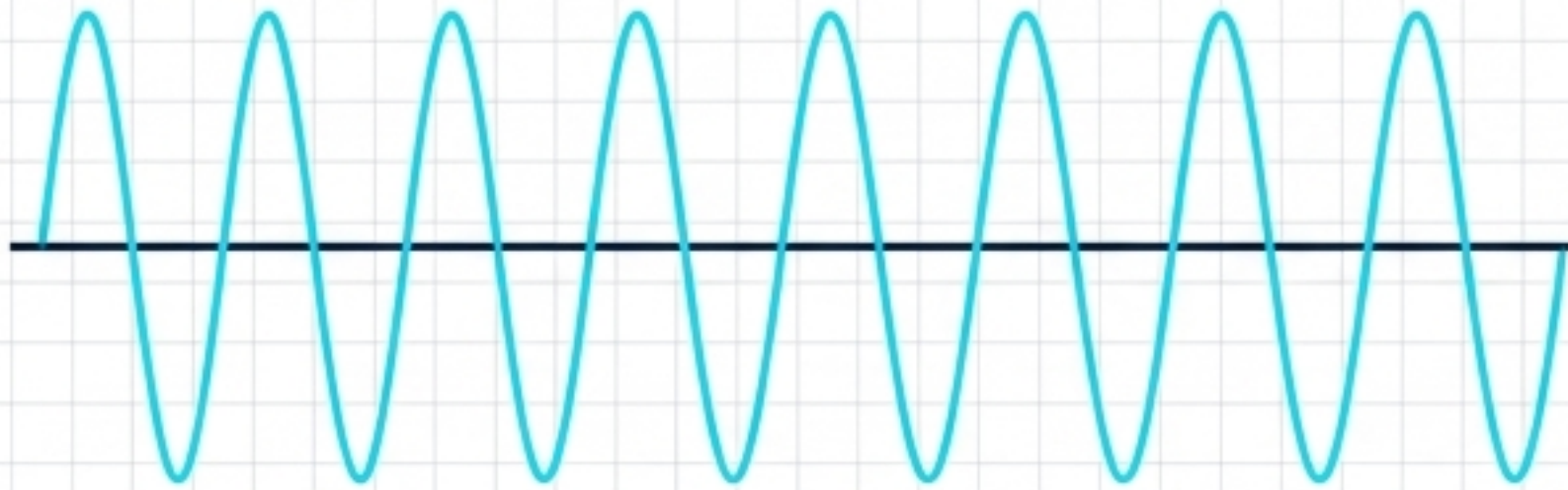
GROWING AMPLITUDE



THE BASELINE RULE: STATIONARITY

Models cannot learn from chaos. The statistical properties of the series must remain constant over time.

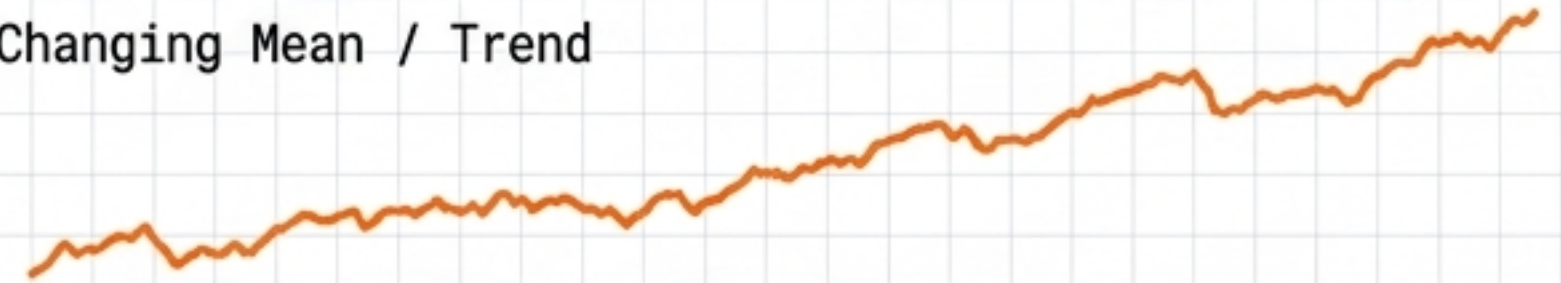
STATIONARY (THE IDEAL)



Metrics: Constant Mean, Constant Variance, Consistent Autocorrelation, No Periodic Components

NON-STATIONARY (THE REALITY)

Changing Mean / Trend



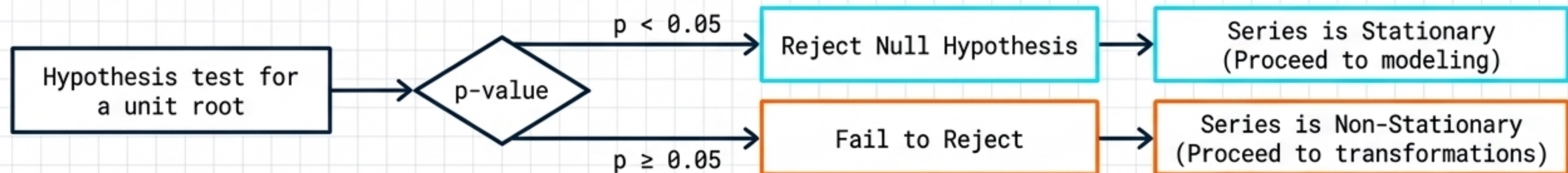
Changing Variance / Heteroscedasticity



Seasonality



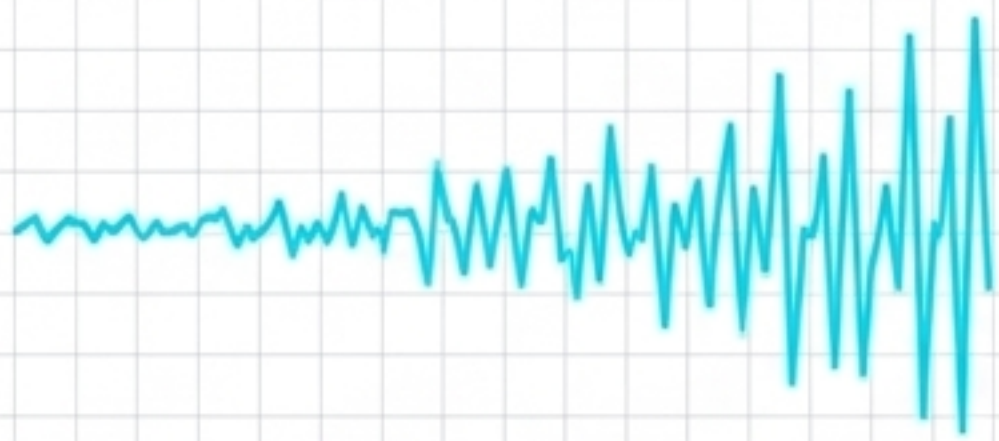
DIAGNOSTIC CHECKPOINT: AUGMENTED DICKEY-FULLER (ADF) TEST



TAMING THE DATA: TRANSFORMATIONS

Treatment protocols for failing the ADF Test.

Symptom: Variance Issue
(Heteroscedasticity)



Treatment: Log Transformation

Mechanism: Squashes variances to make them homoscedastic.

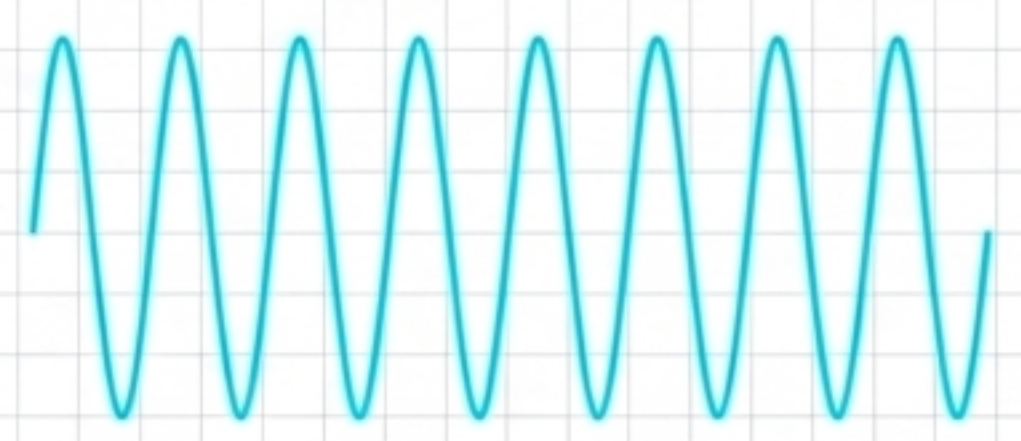
Symptom: Mean /
Autocorrelation Issue



Treatment: Differencing

Mechanism: Subtract the value at a previous time step from the current value ($Value_t - Value_{t-lag}$).

Symptom: Periodic Issue



Treatment: Seasonal Differencing

Mechanism: Subtract values from the exact same season in the previous cycle.



Key Insight: Multiple transformations (e.g., Log followed by Differencing) are often required simultaneously to fully stationarize a raw series.

SMOOTHING THE NOISE

Isolating signal when strict stationarity isn't the primary goal

SIMPLE EXPONENTIAL SMOOTHING

Trigger: Data lacks trend AND lacks seasonality.

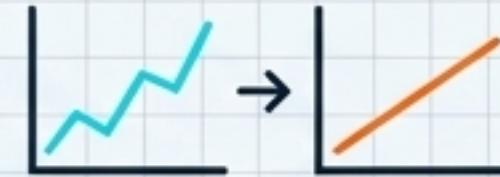
Action: Assigns exponentially decreasing weights to past observations. Captures the level (average) of the series.



DOUBLE EXPONENTIAL SMOOTHING

Trigger: Data has trend, but NO seasonality.

Action: Adds a mathematical component to model continuous upward or downward movement.



TRIPLE / HOLT-WINTERS SMOOTHING

Trigger: Data has BOTH trend AND seasonality.

Action: Models level, trend, and repeating seasonal effects simultaneously.

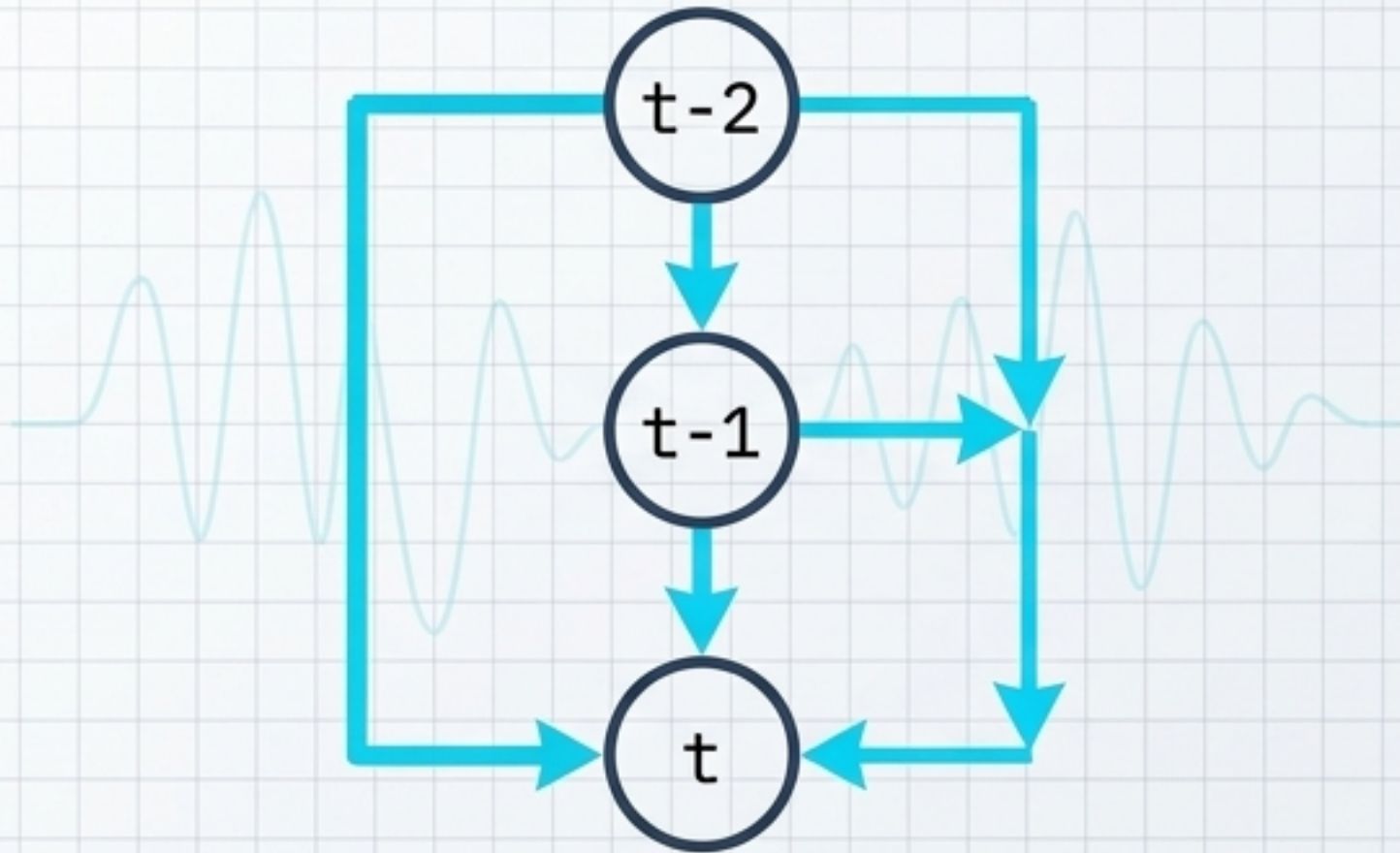


MEMORY & CORRELATION: HOW MODELS 'REMEMBER'

Defining the flow of temporal influence via ACF and PACF.

ACF (Autocorrelation Function)

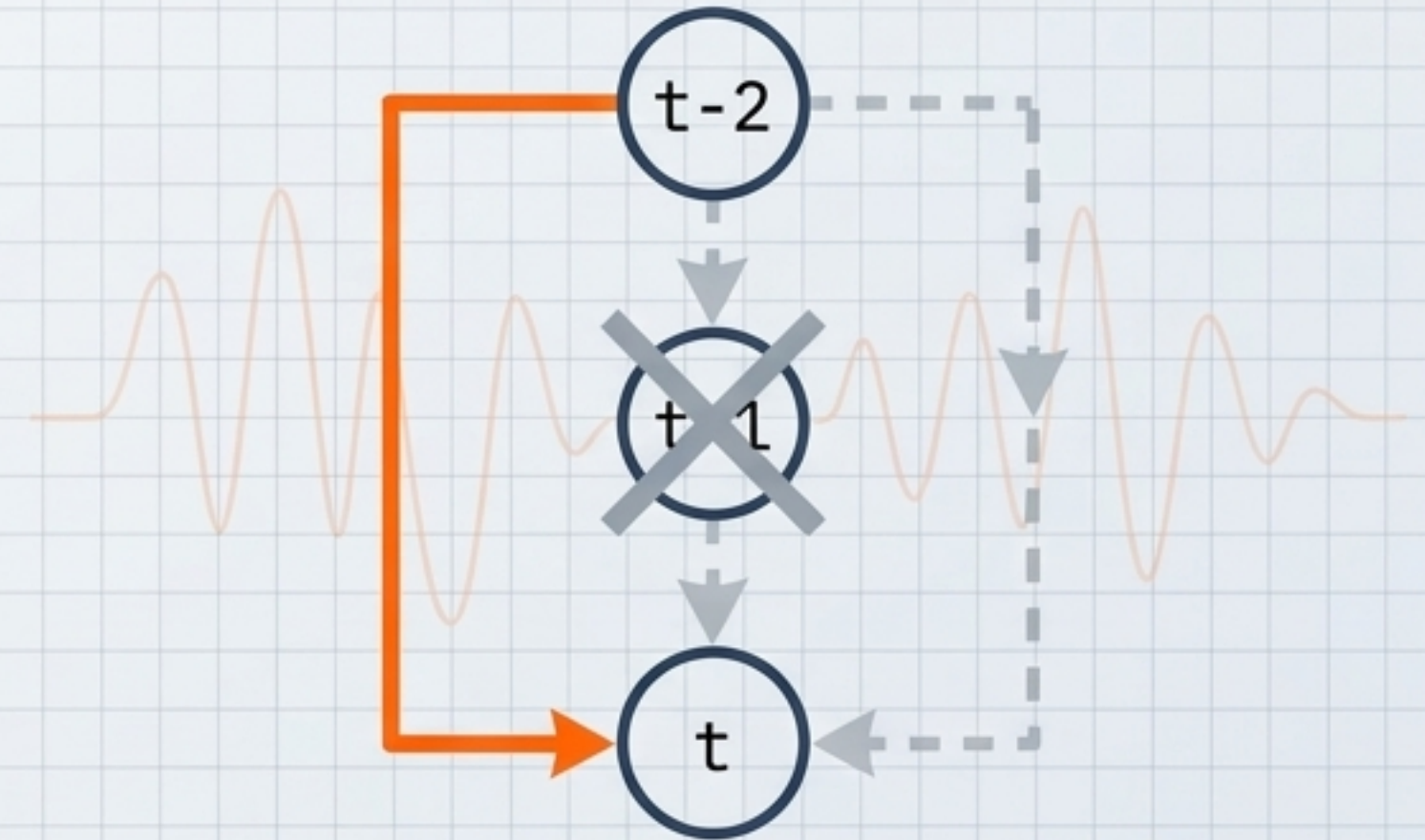
Concept: Total (2-way) correlation between a variable and past values.



Includes all cascading noise and indirect chronological effects.

PACF (Partial Autocorrelation Function)

Concept: Direct correlation at specific lags, controlling for shorter lags.



Isolates the true, unmediated influence of that exact lag.

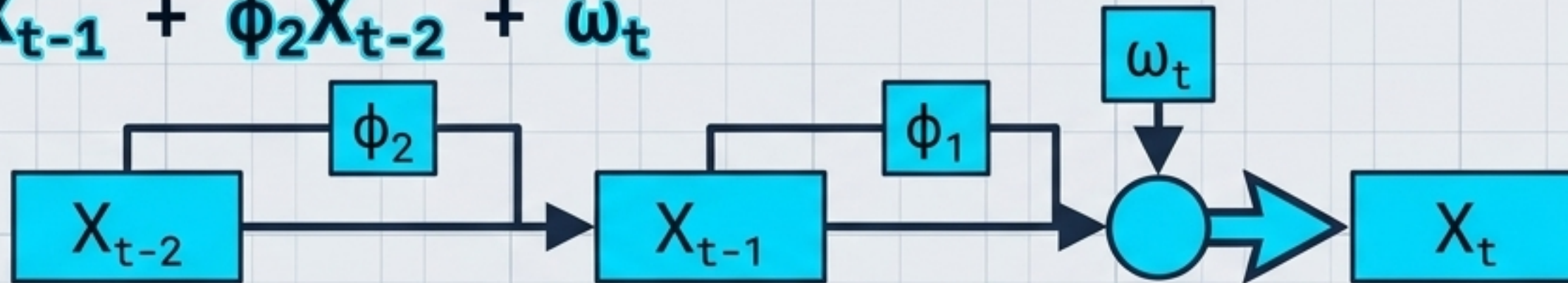
FOUNDATIONAL STATISTICAL MODELS

Deconstructing the two pillars of Box-Jenkins forecasting.

AR (Autoregressive) Models

Concept: Anticipates dependence on its own past values.

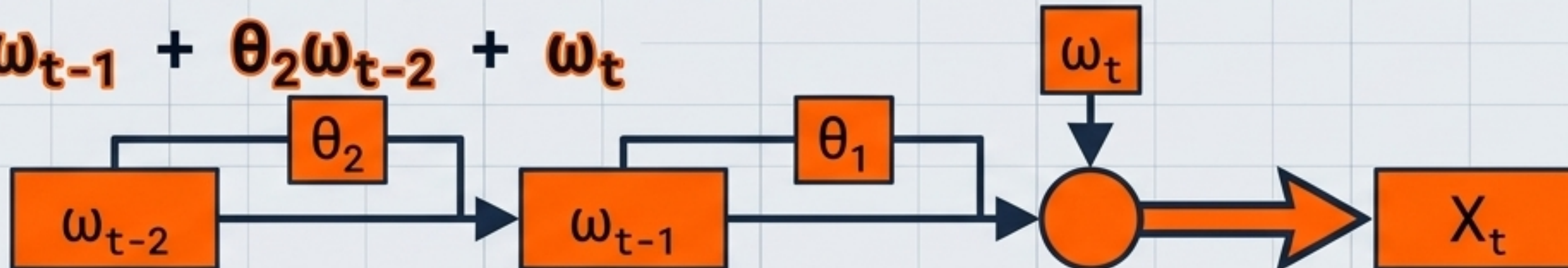
$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \omega_t$$



MA (Moving Average) Models

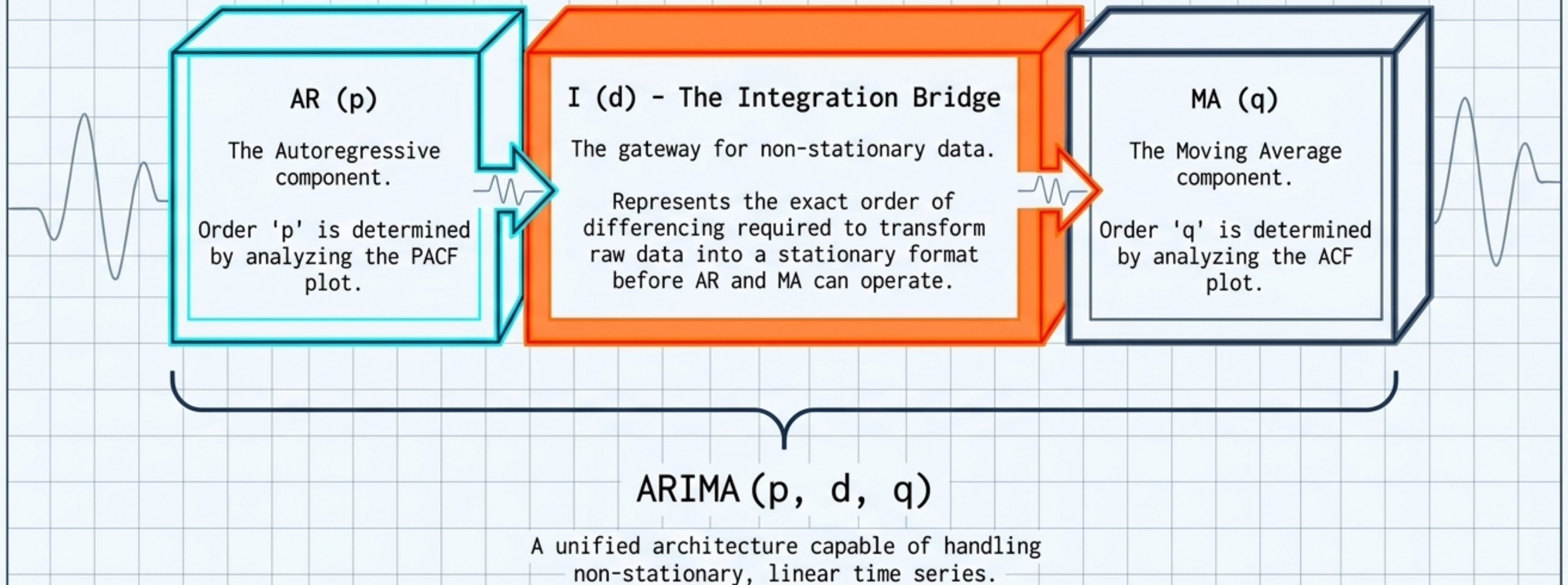
Concept: Anticipates dependence on past forecast errors.

$$X_t = \theta_1 \omega_{t-1} + \theta_2 \omega_{t-2} + \omega_t$$



THE BOX-JENKINS APPROACH: ARIMA

Synthesizing components into a single robust model for linear time series.



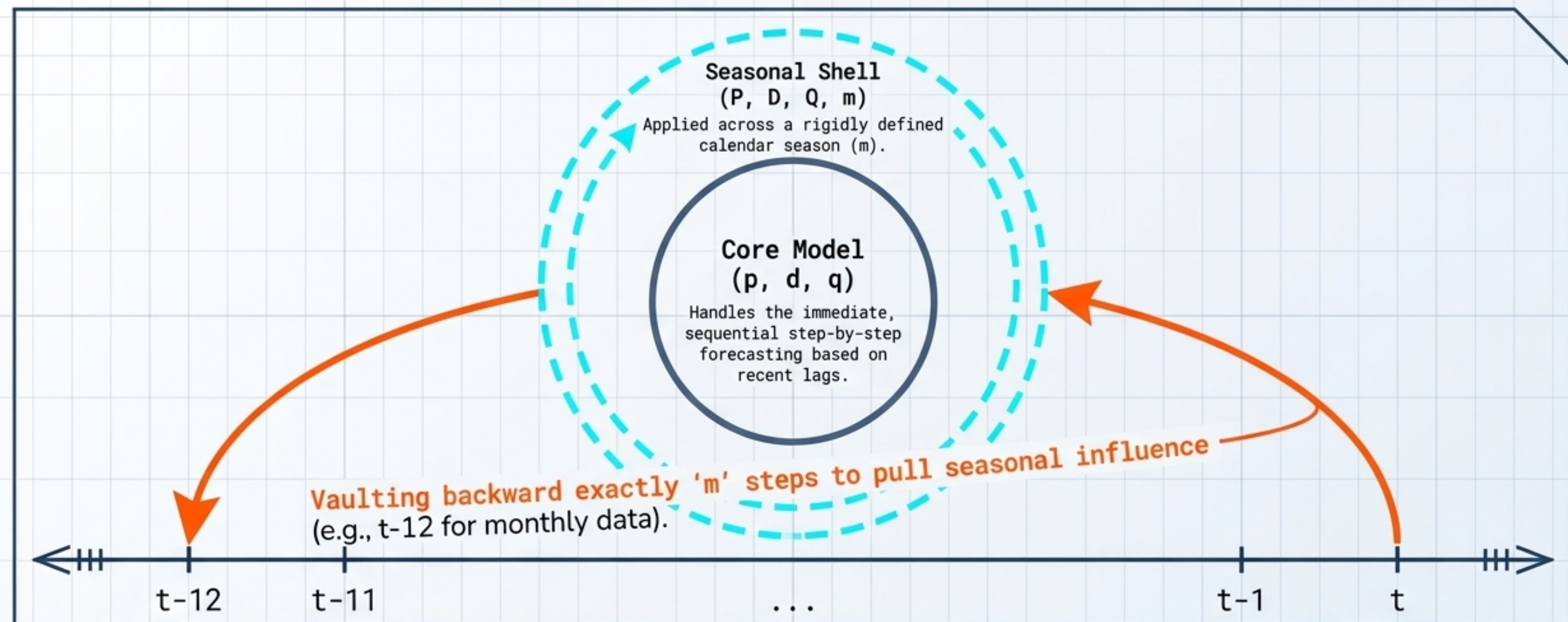
THE ORDER DIAGNOSTIC MATRIX

Translating ACF and PACF visual signatures into precise (p, q) parameters.

SHAPE OF PLOT	RECOMMENDED MODEL ACTION
Exponential decaying to zero OR Alternating positive/negative decaying to zero	AR Models (Use PACF to isolate the exact order p)
One or more sharp spikes, with the rest close to zero	MA Models (Use ACF to isolate the exact order q)
Decay only after a few distinct lags	Mixed ARMA Models (Requires iterative parameter tuning)
High values appearing at fixed, repeating intervals	Include Seasonal Term (Upgrade architecture to SARIMA)
No decay to zero across many lags	STOP. Series remains non-stationary. Return to the differencing phase.

ADDRESSING THE CALENDAR: SARIMA

Leaping over calendar intervals by wrapping ARIMA in a seasonal shell.



Estimation Strategy: Visually inspect run sequence. Generate ACF/PACF. Treat parameters as hyperparameters and optimize via Information Criteria (AIC/BIC) to ruthlessly penalize parameter bloat.

THE DEEP LEARNING FRONTIER: RNNs & LSTMs

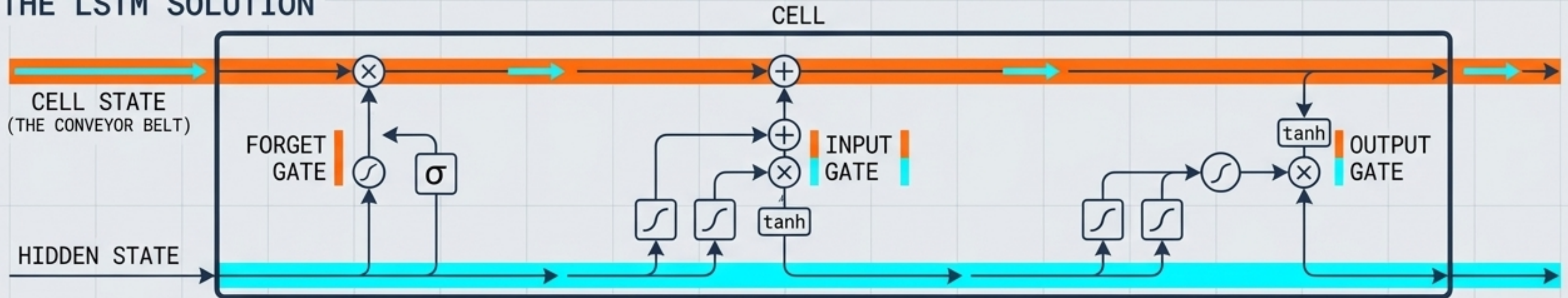
Solving the long-term dependency failure.

THE RNN FAILURE



Mechanism: Information degrades rapidly, failing to retain context across long sequences.

THE LSTM SOLUTION



Mechanism: Gates function as precise valves, regulating information flow to protect memory storage over massive sequence lengths.

TRADE-OFF:

- LSTMs capture complex non-linear patterns.
- REQUIRE MASSIVE DATA, GPUS FOR BACKPROPAGATION.
- Act as highly uninterpretable BLACK BOXES.

MODEL SELECTION MATRIX

Aligning architectural choices with project constraints.

	STATISTICAL MODELS (ARIMA/SARIMA)	DEEP LEARNING (LSTMs)
DATA VOLUME & FEATURES	Small data; strict univariate focus.	Thrives on massive datasets ; easily handles multivariate panels .
PATTERN COMPLEXITY	Linear relationships; explicitly requires stationarity.	Highly non-linear ; automatically learns trend and seasonality internally.
INTERPRETABILITY	Transparent ; mathematically explicitly understood by stakeholders.	Black box ; extremely challenging to explain individual predictions.
COMPUTATIONAL COST	Fast ; runs efficiently on standard CPUs.	Expensive ; slow training via backpropagation; strictly requires GPUs .

AN ALTERNATIVE LENS: SURVIVAL ANALYSIS

Pivoting from forecasting continuous values to forecasting time-to-event.

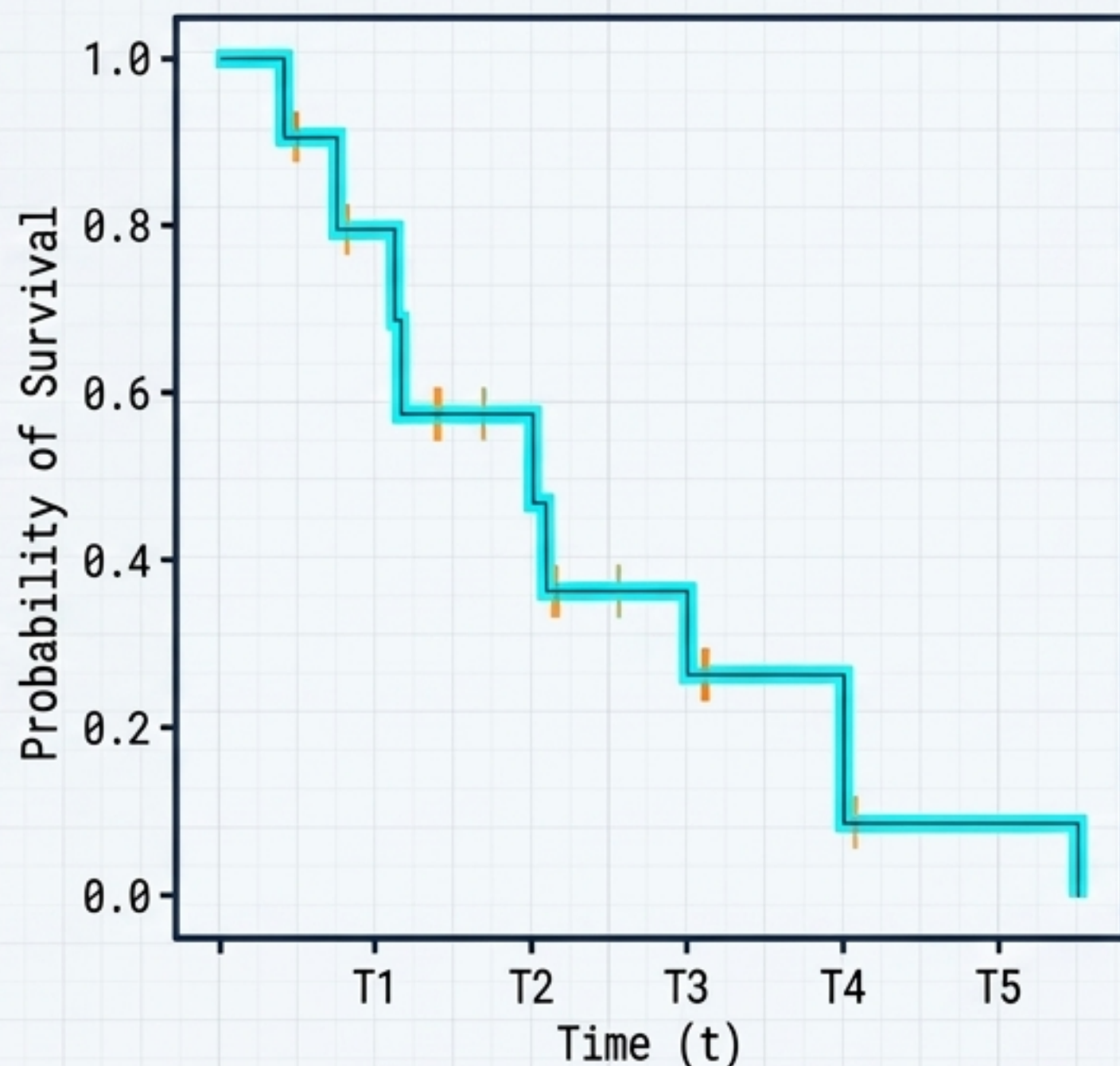
THE SURVIVAL
FUNCTION $S(t)$

$$S(t) = P(T > t)$$

The absolute
probability a subject
survives past time t .

Starts at 1, decreases
strictly to 0.

Masterfully accounts
for incomplete or
censored data.



THE HAZARD RATE $h(t)$

$$h(t) = \frac{f(t)}{S(t)}$$

The instantaneous rate
of an event occurring
(e.g., churn, machine
failure) given it has not
occurred yet.

Cox Proportional Hazard (CPH)

$$h(t) = \beta_0(t)e^{\beta_1 X}$$

Uses features to
proportionally adjust the
baseline hazard rate.

THE FORECASTING LIFECYCLE

A systematic journey from data preparation to model deployment.

